2 Newtonian Dynamics and Gravitation

2.1 Geometric Background of Newton Mechanics

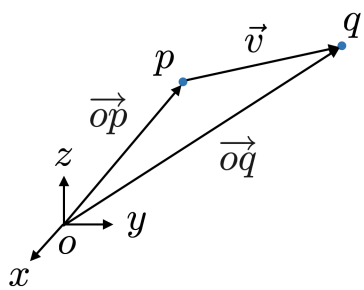

Fig. 2.1: A vector $\vec{v}$ in Euclidean space $\mathbb{R}^3$.

Vector space. In a n-dimensional Euclidean space $\mathbb{R}^n$, we always label a point by coordinates $\{x^i\} = \{x_1, x_2, \cdots, x_n\}$. We can define a point as the origin o with coordinates $(x_1, x_2, \cdots, x_n) = (0, 0, \cdots, 0)$. In addition, any point can be regarded as an end point of an arrow pointed from the origin o. We can naturally use the *Cartesian frame* $\{\vec{e}_1, \vec{e}_2, \cdots, \vec{e}_n\}$ as basis vector to describe such a directional arrow called *position vector*. Now we focus on the three-dimensional Euclidean space $\mathbb{R}^3$. Given two points p and q, shown in Fig. 2.1, and their coordinates are respectively labeled by (x, y, z) and (x', y', z'). We can have position vectors of points p and q written by $\overrightarrow{op} = x\,\vec{e}_x + y\,\vec{e}_y + z\,\vec{e}_z$ and $\overrightarrow{oq} = x'\,\vec{e}_x + y'\,\vec{e}_y + z'\,\vec{e}_z$ respectively.[†1] A *separation vector* $\vec{v}$ is always denoted by an arrow pointed from one point p to another point q. We assume that the separation vector between p and q is $\vec{v} \coloneqq \overrightarrow{pq}$. The separation vector $\vec{v}$ is an arrow pointed from p to q, which can be regarded as a vector represents the action of the movement of a point from p to q. We always call this movement the *displacement vector* at point p. Then, from linear algebra, the vector $\vec{v}$ is the difference of $\overrightarrow{oq}$ and $\overrightarrow{op}$ and we have the relation

$$\begin{aligned}\vec{v} = \overrightarrow{pq} &= \underbrace{\overrightarrow{po}}_{-\overrightarrow{op}} + \overrightarrow{oq} = \overrightarrow{oq} - \overrightarrow{op} \\ &= (x' - x)\,\vec{e}_x + (y' - y)\,\vec{e}_y + (z' - z)\,\vec{e}_z \\ &\equiv v_x\,\vec{e}_x + v_y\,\vec{e}_y + v_z\,\vec{e}_z. \end{aligned} \tag{2.1}$$

Notations of a vector space. Generally we prefer to write a n-dimensional vector by

$$\vec{v} = \sum_i v_i\,\vec{e}_i \coloneqq v_i\,\vec{e}_i \overset{\text{or}}{=} \vec{e}_i\,v_i \quad (i = 1, 2, \ldots, n). \tag{2.2}$$

Here we use the so-called *Einstein's summation convention* in which the repeated index should be realized by the summation over the all components. Consider a three-dimensional vector $\vec{v}$, there are two notations for a vector can be presented here:

[†1] We may sometimes recognize the points p and q by their position vectors denoted by $\mathbf{p} \equiv \overrightarrow{op}$ and $\mathbf{q} \equiv \overrightarrow{oq}$, respectively.

(i) We always use the components (or coordinates) with a *parentheses* or *square bracket* to represent the vector $\vec{v}$ denoted by a *row matrix* (or a *tuple*)

$$\mathsf{v} \equiv (v_i) = \begin{pmatrix} v_x & v_y & v_z \end{pmatrix} \overset{\text{or}}{=} (v_x, v_y, v_z) \quad \text{or} \quad \mathsf{v} \equiv [v_i] = \begin{bmatrix} v_x & v_y & v_z \end{bmatrix} \overset{\text{or}}{=} [v_x, v_y, v_z] \tag{2.3a}$$

or a *column matrix*

$$\mathsf{v} \equiv (v_i) = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \quad \text{or} \quad \mathsf{v} \equiv [v_i] = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}. \tag{2.3b}$$

We note that *matrices* are always denoted by letters in sans serif typeface. Conventionally, people always refer a vector to a column matrix, and such a matrix expression of the vector $\vec{v}$, cf. (2.1), is

$$\vec{v} \longrightarrow \mathsf{v} \equiv (v_i) = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = v_x\, \mathsf{e}_x + v_y\, \mathsf{e}_y + v_z\, \mathsf{e}_z \tag{2.4}$$

by recognizing basis vectors as the following basis matrices

$$\vec{e}_x \equiv \mathsf{e}_x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \vec{e}_y \equiv \mathsf{e}_y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{e}_z \equiv \mathsf{e}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.^{\dagger 2} \tag{2.5}$$

(ii) We always prefer the expression $\vec{v} = \vec{e}_i\, v_i$ (basis times component). The basis vectors are represented by a row matrix, while the corresponding components are represented by a column matrix. So a vector can easily correspond to the multiplication of a row matrix (basis matrix) and a column matrix (component matrix). Define basis and component matrices by

$$\vec{e}_i \longrightarrow \mathsf{e} = \begin{pmatrix} \vec{e}_1 & \vec{e}_2 & \cdots & \vec{e}_n \end{pmatrix}_{1\times n} \quad \text{and} \quad v_i \longrightarrow \mathsf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}_{n\times 1}. \tag{2.6}$$

Therefore, the vector $\vec{v}$ can be understood by the multiplication of matrices e and v:

$$\vec{v} = \vec{e}_i\, v_i \longrightarrow \vec{v} \equiv \mathsf{e} \cdot \mathsf{v}, \quad \text{or more precisely} \quad \vec{v} \equiv \mathsf{e}_{1\times n} \cdot \mathsf{v}_{n\times 1}. \tag{2.7}$$

Inner product. A *bilinear operator* g in a vector space V is introduced such that

$$g(a\,\vec{x}, b\,\vec{y}) = ab\, g(\vec{x}, \vec{y}), \tag{2.8}$$

where $\vec{x}$ and $\vec{y}$ are two vectors in V, and a and b are two real constants. We say that the inner product of two vectors is a bilinear operation of g which can be identified as a function of two *vectors*

$$g(\vec{v}, \vec{w}) = g(\vec{e}_i\, v_i, \vec{e}_j\, w_j) = v_i\, g(\vec{e}_i, \vec{e}_j)\, w_j. \tag{2.9}$$

[†2] This notation is always used as the notation of a *basis ket vector* in *quantum mechanics*.

We can define a matrix

$$\mathsf{G} = (g_{ij}), \tag{2.10}$$

where

$$g_{ij} \coloneqq g(\vec{e}_i, \vec{e}_j) = \vec{e}_i \cdot \vec{e}_j. \tag{2.11}$$

If $\{e_i\}$ is a *normalized* frame, then we would have

$$g_{ij} = \|\vec{e}_i\| \, \|\vec{e}_j\| \, \cos\theta_{ij} = \cos\theta_{ij}. \tag{2.12}$$

where θ_{ij} is the angle between $\vec{e}_i$ and $\vec{e}_j$. Consequently, g_{ij} given in a *normalized* frame is the *direction cosine* of the ith axis relative to the jth axis. Hence, the inner product can be expressed by a multiplication of matrix forms

$$g(\vec{v}, \vec{w}) = v_i \, g_{ij} \, w_j \equiv \mathsf{v}^{\mathrm{T}} \cdot \mathsf{G} \cdot \mathsf{w}, \tag{2.13}$$

where T stands for the *transpose* operation of a matrix and

$$\mathsf{v}^{\mathrm{T}} = \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix}, \quad \mathsf{G} = \begin{pmatrix} g_{11} & g_{12} & \cdots & g_{1n} \\ g_{21} & g_{22} & \cdots & g_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ g_{n1} & g_{n2} & \cdots & g_{nn} \end{pmatrix}, \quad \mathsf{w} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}. \tag{2.14}$$

If $\{\vec{e}_i\}$ is an *orthonormal* frame, then the matrix $g_{ij} = \delta_{ij}$, where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \longrightarrow \quad \mathsf{G} = \operatorname{diag}\big(\underbrace{1, 1, \cdots, 1}_{n \text{ terms}}\big) = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix}_{n \times n} \tag{2.15}$$

is called the *Kronecker delta symbol*. In this situation, we can rewrite the inner product

$$g(\vec{v}, \vec{w}) = v_i \, \delta_{ij} \, w_j \equiv \mathsf{v}^{\mathrm{T}} \cdot \mathsf{G} \cdot \mathsf{w} \tag{2.16a}$$

$$= \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \tag{2.16b}$$

$$= \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \tag{2.16c}$$

$$= v_1 w_1 + v_2 w_2 + \cdots + v_n w_n, \tag{2.16d}$$

which is consist with the ordinary inner product of $\vec{v}$ and $\vec{w}$ in the Euclidean space, i.e., we obtain the identification of

$$\boxed{g(\vec{v}, \vec{w}) = v_i \, g_{ij} \, w_j = \vec{v} \cdot \vec{w},} \tag{2.17}$$

which leads to the norm of a vector $\vec{v}$ by

$$\boxed{\|\vec{v}\| = \sqrt{(v^1)^2 + (v^2)^2 + (v^3)^2}.} \tag{2.18}$$

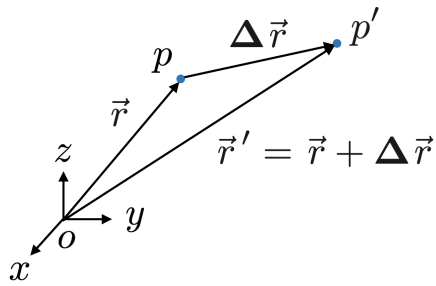

Fig. 2.2: A displacement vector $\boldsymbol{d}\vec{r}$ form the point p to the point q.

The line element. In $\mathbb{R}^3$, we naturally have the position vectors of points p and $p' \equiv q$ are respectively written as $\vec{r} = \vec{e}_x\, x + \vec{e}_y\, y + \vec{e}_z\, z$ and $\vec{r}\,' = \vec{r} + \boldsymbol{\Delta}\vec{r} = \vec{e}_x\, x' + \vec{e}_y\, y' + \vec{e}_z\, z'$, We note that $\boldsymbol{\Delta}\vec{r}$ is a displacement vector from p to q.[†3] This is shown in Fig. 2.2. To emphasize the *fixed* Cartesian frame, we sometime denote it by the notation of $\{\vec{e}_i\} = \{\vec{\delta}_i\}$ for $i = x, y, z$. Thus the displacement vector in *Cartesian coordinates* is usually read by

$$\boldsymbol{\Delta}\vec{r} = \overrightarrow{pq} = \overrightarrow{oq} - \overrightarrow{op} = \vec{r}\,' - \vec{r} \tag{2.19a}$$

$$= \vec{e}_x\, \Delta x + \vec{e}_y\, \Delta y + \vec{e}_z\, \Delta z \quad (\overset{\text{or}}{=} \vec{\delta}_i\, \Delta x^i). \tag{2.19b}$$

We note that $\boldsymbol{\Delta}$ in **bold-face** is used to denote the *variation of a vector* (a *vectorial* variation) in order to distinguish the ordinary variation of a function, which includes not only the difference of a function but also the addition law of vectors in linear algebra. The variation of a displacement vector is realized by

$$\begin{aligned}\boldsymbol{\Delta}\vec{r} &= \boldsymbol{\Delta}(\vec{e}_x\, x + \vec{e}_y\, y + \vec{e}_z\, z) \\ &= \underbrace{\boldsymbol{\Delta}\vec{e}_x}_{0}\, x + \vec{e}_x\, \Delta x + \underbrace{\boldsymbol{\Delta}\vec{e}_y}_{0}\, y + \vec{e}_y\, \Delta y + \underbrace{\boldsymbol{\Delta}\vec{e}_z}_{0}\, z + \vec{e}_z\, \Delta z.\end{aligned} \tag{2.20}$$

Therefore, due to the *invariant* basis of the Cartesian frame $\{\vec{e}_i\}$, the result obtained coincides with (2.19). The norm of the displacement vector $\boldsymbol{\Delta}\vec{r}$ is denoted by ΔS, which can be obtained by the inner product:

$$\Delta S := \|\boldsymbol{\Delta}\vec{r}\| = \sqrt{\boldsymbol{\Delta}\vec{r}\cdot\boldsymbol{\Delta}\vec{r}} = \sqrt{g(\boldsymbol{\Delta}\vec{r}, \boldsymbol{\Delta}\vec{r})}. \tag{2.21}$$

We define the *line element* of a space as the square of the norm of $\boldsymbol{\Delta}\vec{r}$ by

$$\Delta S^2 = \boldsymbol{\Delta}\vec{r}\cdot\boldsymbol{\Delta}\vec{r} = g\big(\boldsymbol{\Delta}\vec{r}, \boldsymbol{\Delta}\vec{r}\big) = g\big(\vec{e}_i\, \Delta x_i, \vec{e}_j\, \Delta x_j\big) = g\big(\vec{e}_i, \vec{e}_j\big)\Delta x_i \Delta x_j = g_{ij}\, \Delta x_i \Delta x_j. \tag{2.22}$$

In physics people are always interested in the *infinitesimal* form. If two points p and q are separated by an *infinitesimal distance* dS, then we have the *vectorial differential* of the position vector

$$\boldsymbol{\Delta}\vec{r} \quad\longrightarrow\quad \boldsymbol{d}\vec{r} = \overrightarrow{pq} = \overrightarrow{oq} - \overrightarrow{op} = \vec{r}\,' - \vec{r} \tag{2.23a}$$

$$= \vec{e}_x\, dx + \vec{e}_y\, dy + \vec{e}_z\, dz \quad (\overset{\text{or}}{=} \delta_i\, dx_i). \tag{2.23b}$$

[†3] Generally we can denote the position vector of the point p by $\mathbf{p}$ (or by a matrix expression p) and write the displacement vector by $\boldsymbol{\Delta}\mathbf{p} \equiv \mathbf{p}' - \mathbf{p}$ (or by a matrix expression $\boldsymbol{\Delta}\mathsf{p}$) or simply $\boldsymbol{\Delta}p$. Correspondingly, its infinitesimal form should be $\boldsymbol{d}\mathbf{p}$ (or by a matrix expression $\boldsymbol{d}\mathsf{p}$) or simply $\boldsymbol{d}p$ respectively.

So the *line element* of a space becomes

$$dS^2 = \boldsymbol{d}\vec{r} \cdot \boldsymbol{d}\vec{r} = g\big(\boldsymbol{d}\vec{r}, \boldsymbol{d}\vec{r}\big) = g_{ij}\, dx_i dx_j. \tag{2.24}$$

For two-dimensional Euclidean space, the displacement vector is $\boldsymbol{d}\vec{r} = \vec{e}_x\, dx + \vec{e}_y\, dy = \vec{e}_i\, dx_i$ and the corresponding line element with $g_{ij} = \delta_{ij}$ is

$$dS^2 = \delta_{ij}\, dx_i dx_j \tag{2.25a}$$

$$\equiv \begin{pmatrix} dx & dy \end{pmatrix} \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix}. \tag{2.25b}$$

Similarly, the three-dimensional line element with $g_{ij} = \delta_{ij}$ and $\boldsymbol{d}\vec{r} = \vec{e}_x\, dx + \vec{e}_y\, dy + \vec{e}_z\, dz = \vec{e}_i\, dx_i$ is

$$dS^2 = \delta_{ij}\, dx_i dx_j \tag{2.26a}$$

$$\equiv \begin{pmatrix} dx & dy & dz \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix}. \tag{2.26b}$$

Example. (Curvilinear Coordinates).

In stead of using the Cartesian coordinates, we can choose *curvilinear coordinates* for calculations. We would like to know the displacement vector $\boldsymbol{d}\vec{r} = \vec{r}\,' - \vec{r}$ and line element in these coordinate systems.

Polar coordinates in $\mathbb{R}^2$.

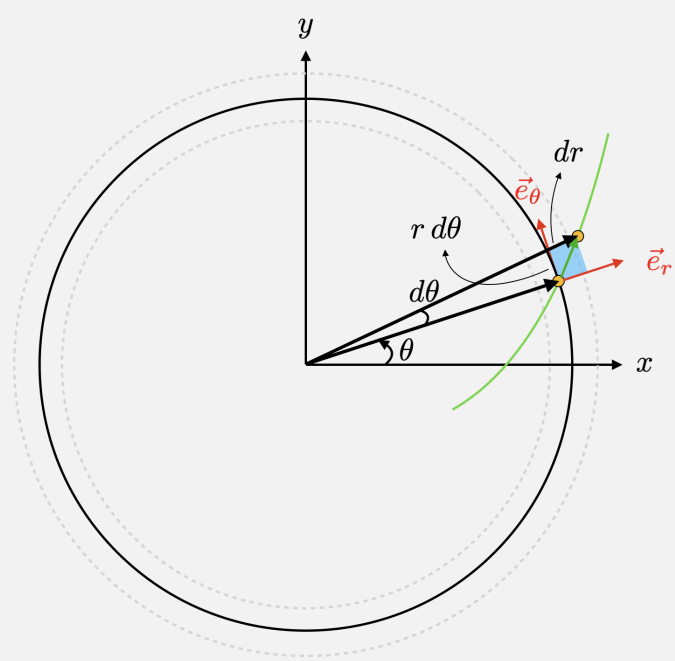

Fig. 2.3: Polar coordinates in $\mathbb{R}^2$.

Given a position vector

$$\vec{r}(x, y) = \vec{e}_x\, x + \vec{e}_y\, y, \tag{2.27}$$

where the coordinates (x, y) in terms of the polar coordinates are

$$\begin{cases} x = r\cos\theta, & (2.28a) \\ y = r\sin\theta. & (2.28b) \end{cases}$$

So we would like to consider the coordinate transformation

$$(x, y) \quad \longrightarrow \quad (r, \theta). \tag{2.29}$$

What about the coordinate transformation of the basis vector? That means we will have

$$\{\vec{e}_x, \vec{e}_y\} \quad \longrightarrow \quad \{\vec{e}_r, \vec{e}_\theta\}. \tag{2.30}$$

We know that basis vectors should be the tangent vectors along the corresponding direction of coordinates. Hence, we can obtain basis vectors through the derivatives with respect to the orresponding coordinates

$$\begin{aligned} \vec{V}_r = \frac{\partial \vec{r}}{\partial r} &= \frac{\partial(\vec{e}_x\, x + \vec{e}_y\, y)}{\partial r} \\ &= \underbrace{\frac{\partial \vec{e}_x}{\partial r}}_{0}\, r\cos\theta + \vec{e}_x\, \frac{\partial(r\cos\theta)}{\partial r} + \underbrace{\frac{\partial \vec{e}_y}{\partial r}}_{0}\, r\sin\theta + \vec{e}_y\, \frac{\partial(r\sin\theta)}{\partial r} \\ &= \vec{e}_x\cos\theta + \vec{e}_y\sin\theta \quad \longrightarrow \quad \boxed{\vec{e}_r := \frac{\vec{V}_r}{\|\vec{V}_r\|} = \vec{e}_x\cos\theta + \vec{e}_y\sin\theta,} \end{aligned} \tag{2.31a}$$

$$\begin{aligned} \vec{V}_\theta = \frac{\partial \vec{r}}{\partial \theta} &= \frac{\partial(\vec{e}_x\, x + \vec{e}_y\, y)}{\partial r} \\ &= \vec{e}_x\, \frac{\partial(r\cos\theta)}{\partial \theta} + \vec{e}_y\, \frac{\partial(r\sin\theta)}{\partial \theta} \\ &= -\vec{e}_x\, r\sin\theta + \vec{e}_y\, r\cos\theta \quad \longrightarrow \quad \boxed{\vec{e}_\theta := \frac{\vec{V}_\theta}{\|\vec{V}_\theta\|} = -\vec{e}_x\sin\theta + \vec{e}_y\cos\theta,} \end{aligned} \tag{2.31b}$$

where the *scale factors* are defined as the *norm* of tangent vectors

$$\|\vec{V}_r\| = 1 := h_r, \tag{2.32a}$$

$$\|\vec{V}_\theta\| = r := h_\theta. \tag{2.32b}$$

The position vector in the polar coordinates is

$$\boxed{\begin{aligned} & \vec{r} = \vec{e}_x(r\cos\theta) + \vec{e}_y(r\sin\theta) = \vec{e}_r\, r \\ \Longrightarrow\ & \text{No cooordinate } \theta \text{ in the position vector.} \\ \Longrightarrow\ & \text{We would have a } \textit{symmetry} \text{ in the } \theta\text{-direction.} \end{aligned}} \tag{2.33}$$

By differentiating the position vector, the displacement vector can be regarded as a total differential of the vector $\vec{r}$:

$$\begin{aligned}
\boldsymbol{d}\vec{r} &= \boldsymbol{d}\big(\vec{e}_x\, x + \vec{e}_y\, y\big) \\
&= \underbrace{\boldsymbol{d}\vec{e}_x}_{0}(r\,\cos\theta) + \vec{e}_x\,\boldsymbol{d}(r\,\cos\theta) + \underbrace{\boldsymbol{d}\vec{e}_y}_{0}(r\,\sin\theta) + \vec{e}_y\,\boldsymbol{d}(r\,\sin\theta) \\
&= \vec{e}_x(dr\,\cos\theta - r\,\sin\theta\,d\theta) + \vec{e}_y(dr\,\sin\theta + r\,\cos\theta\,d\theta) \\
&= \underbrace{(\vec{e}_x\,\cos\theta + \vec{e}_y\,\sin\theta)}_{\vec{e}_r\,h_r = \vec{V}_r = \frac{\partial\vec{r}}{\partial r}}\,dr + \underbrace{(-\vec{e}_x\,\sin\theta + \vec{e}_y\,\cos\theta)\,r}_{\vec{e}_\theta\,h_\theta = \vec{V}_\theta = \frac{\partial\vec{r}}{\partial\theta}}\,d\theta.
\end{aligned} \tag{2.34}$$

On the other hand, the displacement vector can be directly realized in the polar coordinates by

$$\boldsymbol{d}\vec{r} = \boldsymbol{d}(\vec{e}_r\,r) = \boldsymbol{d}\vec{e}_r + \vec{e}_r\,dr. \tag{2.35}$$

By comparing (2.34) and (2.35), we have found that

$$\boxed{\boldsymbol{d}\vec{e}_r = \vec{e}_\theta\,d\theta.} \tag{2.36}$$

Alternatively, we can use (2.31) respectively to calculate $\boldsymbol{d}\vec{e}_r$ and $\boldsymbol{d}\vec{e}_\theta$:

$$\begin{cases}
\boldsymbol{d}\vec{e}_r = -\vec{e}_x\sin\theta\,d\theta + \vec{e}_y\cos\theta\,d\theta = +\vec{e}_\theta\,d\theta, & \text{(2.37a)} \\
\boldsymbol{d}\vec{e}_\theta = -\vec{e}_x\cos\theta\,d\theta - \vec{e}_y\sin\theta\,d\theta = -\vec{e}_r\,d\theta. & \text{(2.37b)}
\end{cases}$$

Hence, it results in that the displacement vector in the polar coordinates is given by

$$\boxed{\boldsymbol{d}\vec{r} = \frac{\partial\vec{r}}{\partial r}\,dr + \frac{\partial\vec{r}}{\partial\theta}\,d\theta = \vec{V}_r\,dr + \vec{V}_\theta\,d\theta \equiv \vec{e}_r(h_r\,dr) + \vec{e}_\theta(h_\theta\,d\theta),} \tag{2.38}$$

the sets $\{\vec{V}_r, \vec{V}_\theta\}$ and $\{\vec{e}_r, \vec{e}_\theta\}$ are both basis vector of the tangent space. The *line element* in coordinates (r,θ) is defined as the square of the norm of the displacement vector $\boldsymbol{d}\vec{r}$:

$$\begin{aligned}
dS^2 = g(\boldsymbol{d}\vec{r}, \boldsymbol{d}\vec{r}) &= h_r^2\,dr^2 + h_\theta^2\,d\theta^2 \\
&\equiv \begin{pmatrix} h_r\,d\rho & h_\theta\,d\theta \end{pmatrix} \begin{pmatrix} h_r\,dr \\ h_\theta\,d\theta \end{pmatrix} \\
&= \begin{pmatrix} h_r\,d\rho & h_\theta\,d\theta \end{pmatrix} \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \begin{pmatrix} h_r\,dr \\ h_\theta\,d\theta \end{pmatrix} \\
&\overset{\text{or}}{=} \begin{pmatrix} d\rho & d\theta \end{pmatrix} \underbrace{\begin{pmatrix} h_r^2 & \\ & h_\theta^2 \end{pmatrix}}_{g_{ij}} \begin{pmatrix} dr \\ d\theta \end{pmatrix}
\end{aligned} \tag{2.39}$$

with the metric tensor is

$$g_{ij} = \vec{V}_i \cdot \vec{V}_j = \begin{pmatrix} \vec{V}_r \cdot \vec{V}_r & \\ & \vec{V}_\theta \cdot \vec{V}_\theta \end{pmatrix} \equiv \begin{pmatrix} h_r^2 & \\ & h_\theta^2 \end{pmatrix} = \begin{pmatrix} 1 & \\ & r^2 \end{pmatrix} \quad (i,j = r,\theta). \tag{2.40}$$

In most situations, we would like to know the motion of a particle described, i.e., the trajectory of a particle, which is represented by the position vector $\vec{r}$. We needs to solve the equation of motion to and then obtain the solution $\vec{r}$. Therefore, it is hard to know the symmetry by reading the information from $\vec{r}$ initially, such as what we did in (2.33). However, the *displacement vector*(2.38) or the *line element* (2.39) contains most information of the geometric structure, we can start to calculate what we want, e.g., the equation of motion, we will explain in the following discussions, from one of them, and finally find the solution of $\vec{r}$.

Cylindrical coordinates in $\mathbb{R}^3$.

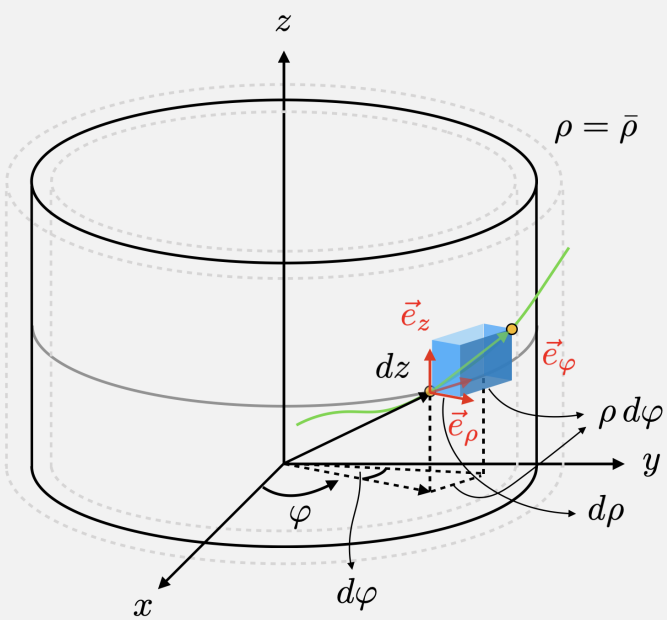

Fig. 2.4: Cylindrical coordinates in $\mathbb{R}^3$.

The coordinates (x, y, z) in terms of the cylindrical coordinates are

$$\begin{cases} x = \rho\cos\varphi, & (2.41a) \\ y = \rho\sin\varphi, & (2.41b) \\ z = z. & (2.41c) \end{cases}$$

Hence, the basis vectors should be

$$\begin{cases} \vec{e}_\rho = \vec{e}_x\cos\varphi + \vec{e}_y\sin\varphi, & (2.42a) \\ \vec{e}_\varphi = -\vec{e}_x\sin\varphi + \vec{e}_y\cos\varphi, & (2.42b) \\ \vec{e}_z = \vec{e}_z. & (2.42c) \end{cases}$$

The position vector is

$$\boxed{\begin{array}{ll} & \vec{r} = \vec{\rho} + \vec{z} = \vec{e}_\rho\,\rho + \vec{e}_z\,z \\ \Longrightarrow & \text{The } \varphi\text{-direction provides symmetry.} \end{array}}$$

The displacement vector

$$\begin{aligned}\boldsymbol{d}\vec{r} = \boldsymbol{d}(\vec{e}_\rho\,\rho + \vec{e}_z\,z) &= \underbrace{\boldsymbol{d}\vec{e}_\rho}_{\vec{e}_\varphi\,d\varphi}\,\rho + \vec{e}_\rho\,d\rho + \underbrace{\boldsymbol{d}\vec{e}_z}_{0}\,z + \vec{e}_z\,dz \\ &= \vec{e}_\rho\,d\rho + \vec{e}_\varphi(\rho\,d\varphi) + \vec{e}_z\,dz. \end{aligned} \tag{2.43}$$

The line element in (ρ, φ, z) can be written by

$$\begin{aligned} dS^2 &= d\rho^2 + \rho^2\,d\varphi^2 + dz^2 \\ &\equiv \begin{pmatrix} d\rho & d\varphi & dz \end{pmatrix} \underbrace{\begin{pmatrix} 1 & & \\ & \rho^2 & \\ & & 1 \end{pmatrix}}_{g_{ij}} \begin{pmatrix} d\rho \\ d\varphi \\ dz \end{pmatrix}. \end{aligned} \tag{2.44}$$

Spherical coordinates in $\mathbb{R}^3$.

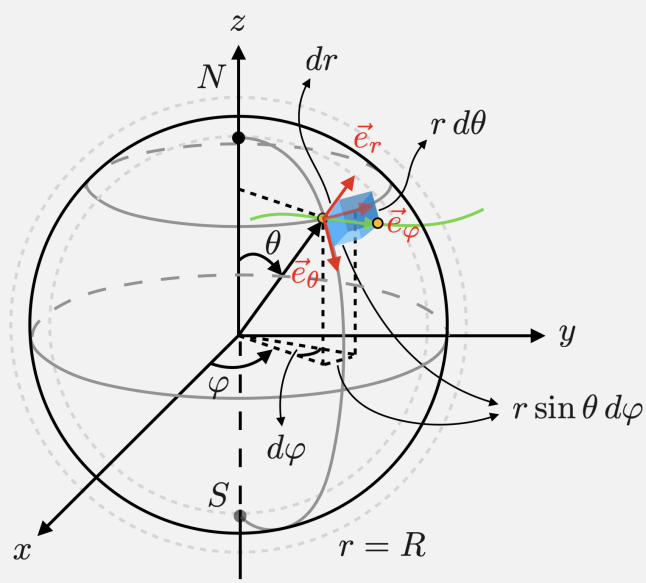

Fig. 2.5: Spherical coordinates in $\mathbb{R}^3$.

The coordinates (x, y, z) in terms of the spherical coordinates are

$$\begin{cases} x = r\sin\theta\cos\varphi, & (2.45a) \\ y = r\sin\theta\sin\varphi, & (2.45b) \\ z = r\cos\theta. & (2.45c) \end{cases}$$

Hence, the basis vectors should be

$$\begin{cases} \vec{e}_r = \vec{e}_x\sin\theta\cos\varphi + \vec{e}_y\sin\theta\sin\varphi + \vec{e}_z\cos\theta, & (2.46a) \\ \vec{e}_\theta = \vec{e}_x\cos\theta\cos\varphi + \vec{e}_y\cos\theta\sin\varphi - \vec{e}_z\sin\theta, & (2.46b) \\ \vec{e}_\varphi = -\vec{e}_x\sin\varphi + \vec{e}_y\cos\varphi. & (2.46c) \end{cases}$$

The position vector in the spherical coordinates is

$$\boxed{\begin{aligned} & \vec{r} = \vec{e}_r\, r \\ \Longrightarrow\quad & \text{There exists a symmetry on the } (\theta, \varphi)\text{-surface.} \end{aligned}}$$

The displacement vector is

$$\boldsymbol{d}\vec{r} = \boldsymbol{d}(\vec{e}_r\, r) = \underbrace{\boldsymbol{d}\vec{e}_r}_{\vec{e}_\theta\, d\theta + \vec{e}_\varphi(\sin\theta\, d\varphi)}\, r + \vec{e}_r\, dr = \vec{e}_r\, dr + \vec{e}_\theta(r\, d\theta) + \vec{e}_\varphi(r\, \sin\theta\, d\varphi). \tag{2.47}$$

The line element in (r, θ, φ) can be written by

$$\begin{aligned} dS^2 &= dr^2 + r^2\, d\theta^2 + r^2\, \sin^2\theta\, d\varphi^2 \\ &\equiv \begin{pmatrix} dr & d\theta & d\varphi \end{pmatrix} \underbrace{\begin{pmatrix} 1 & & \\ & r^2 & \\ & & r^2 \sin^2\theta \end{pmatrix}}_{g_{ij}} \begin{pmatrix} dr \\ d\theta \\ d\varphi \end{pmatrix}. \end{aligned} \tag{2.48}$$

2.2 Newton's Force Equation

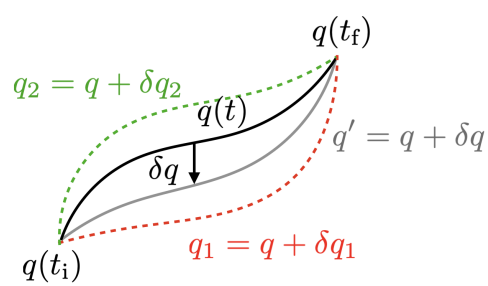

Fig. 2.6: Possible trajectories of a particle.

We first consider, for simplicity, that the space is a *one-dimensional Euclidean space* $\mathbb{R}$ and then a configuration space spanned by the *trajectory* (or called *history*) $q(t)$ and the *velocity* $\dot{q}(t)$ of a point particle, where $q(t)$ is the generalized coordinate parameterized by the time parameter t. We assume that the point particle carries mass with a unit, i.e., $m = 1$, then the *kinetic energy* $T(\dot{q})$ can be read as

$$T(\dot{q}) = \frac{1}{2}\dot{q}^2, \tag{2.49}$$

where $\dot{q} \coloneqq dq/dt$. The *potential energy* is also simply given by $U(q)$ as a function of coordinate only. Then we can define the *Lagrangian function* (or *Lagrangian*) $L(q, \dot{q}; t)$ which is implicitly t-dependence

$$L(q, \dot{q}; t) = T(\dot{q}) - U(q). \tag{2.50}$$

The corresponding *conjugate momentum* of the generalized coordinate q is

$$p(t) \coloneqq \frac{\partial L(q, \dot{q})}{\partial \dot{q}} = \frac{dT(\dot{q})}{d\dot{q}}. \tag{2.51}$$